Strategy 3: Using Jumps in Prediction Market Probabilities as a Signal

Research Question

Is there alpha in using prediction market probabilities as a predictive signal for cryptocurrency price movement?

- Do sharp jumps in probabilities in relevant markets lead crypto movement?
- How would we define a “jump” and are these signals tradeable?
- How does this compare to traditional sentiment strategies?

Literature Review

Paper by NBER and JHU analyzed Kalshi markets for CPI, unemployment, current affairs, etc.
Relevant findings:

- Contract prices reflect accurate real-time probabilities of future events
- Captures full distributions (mean, variance, skew, volatility) rather than just point estimates
- Market probabilities are comparable to or better than surveys or futures
- Reacts immediately to macro events and news

Hypothesis: Crypto markets may take longer to incorporate market beliefs compared to prediction markets, could there be an event driven strategy here?

Data

- Kalshi 1m data for relevant markets
- Binance BTCUSDT, ETHUSDT, SOLUSDT 1m data
- Sentiment data via LunarCrush API for BTC, ETH
- DVOL index (BTC, ETH volatility) - for extra analysis

Market Selection - Categorization

01	Price related markets	• Will BTC/ETH/SOL hit $95k by Dec 2026? • How high will ETH go this year? • When will Bitcoin hit $120k?
02	Regulatory/Institutional	• ETF Approval • Which stablecoins will get GENIUS Act approval? • When will crypto market structure legislation become law?
03	Macro Markets	• GDP/Employment • Interest rates cuts • CPI

Market Selection

- Included as many markets as possible and then ran automated keyword search

Ran current universe through filters:

- History: ≥ 30 active days
- Volume: ≥ $3000 - 5000 daily (avg)
- Spread: ≤ 6-12% bid-ask spread
- Data density: 1m resolution available

```
KXBTCMAX100-26-DEC [T1] PASS ✓ days= 50 vol=  3,702 spread=  3% oi=  66,223
KXBTCMAX100-26-SEP [T1] PASS ✓ days= 50 vol=  3,513 spread=  5% oi=  79,153
KXBTCMAX100-26-MAY [T1] PASS ✓ days= 93 vol=  6,856 spread=  8% oi= 283,566
KXBTCMAX100-26-MAR [T1] PASS ✓ days= 87 vol= 15,226 spread= 51% oi= 772,709
KXBTCMAX100-26-JUNE [T1] PASS ✓ days= 93 vol=  8,406 spread=  7% oi= 347,196
KXBTCMAX100-26-JAN [T1] FAIL ✗ days= 28 vol= 14,610 spread= 38% oi= 143,904 (days=28<30)
KXBTCMAX100-26-FEB [T1] PASS ✓ days= 56 vol= 12,135 spread= 58% oi= 370,511
KXBTCMAX100-26-APR [T1] PASS ✓ days= 93 vol= 11,593 spread= 19% oi= 491,929
```

53 Markets

Defining “spikes” in Probabilities

Rolling Z-Score

Detects abnormal changes based on recent history. Considers rolling mean and volatility

Threshold

Fires when $\Delta P > X$. Testing for X = 0.05, 0.10, 0.15

Percentage Change

Activated when ΔP > % of current probability level. Testing for 20%

Rolling Z-Score with liquidity

Same as 1 but with a volume filter. Less signals but potentially more accurate

EDA

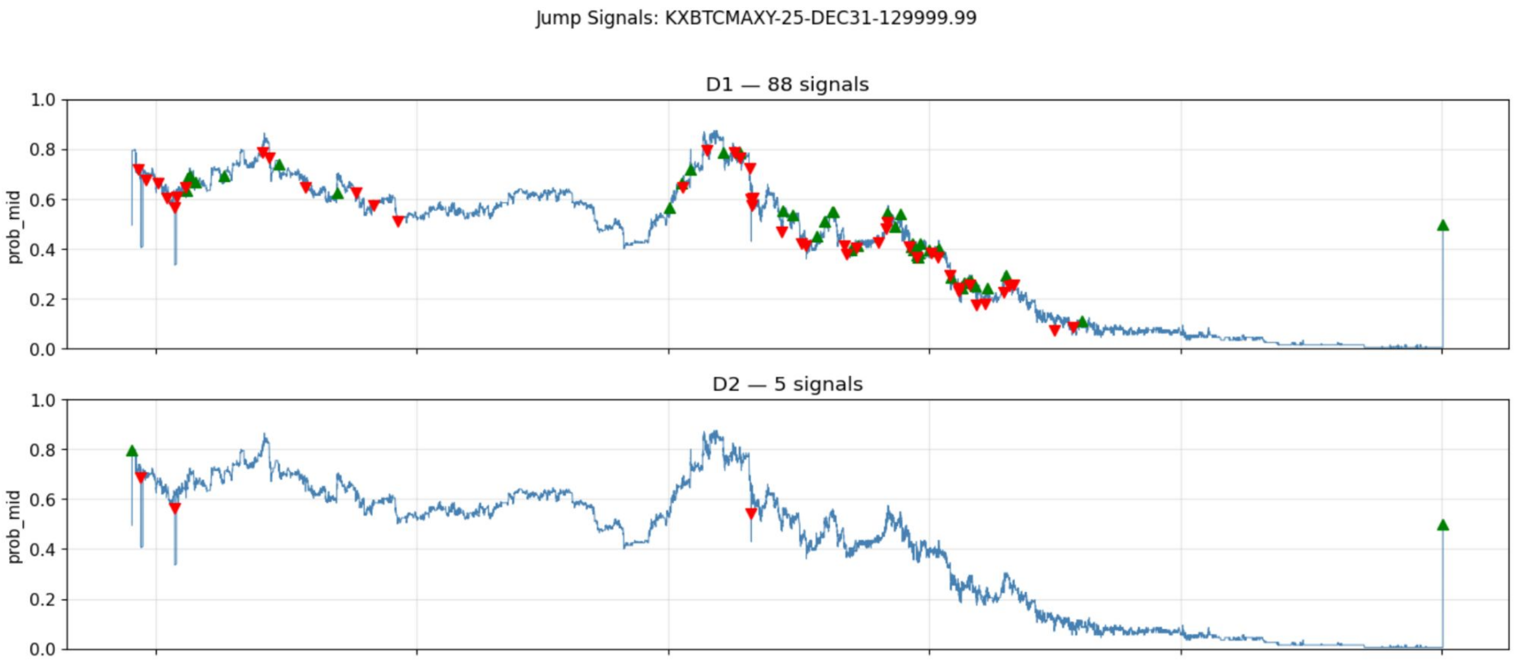
Total Signal Counts:
- D1: 2293
- D2: 618
- D3: 874
- D4: 1323

60% of the markets have more than 50 signals - ignoring the 40% in lead-lag analysis

Signal Count for Each Strategy

definition	D1	D2	D3	D4
market				
KXBTCMAX100-26-APR	54	12	27	30
KXBTCMAX100-26-DEC	26	1	1	13
KXBTCMAX100-26-FEB	76	14	40	45
KXBTCMAX100-26-JUNE	106	15	9	51
KXBTCMAX100-26-MAR	60	11	35	31
KXBTCMAX100-26-MAY	61	14	4	26
KXBTCMAX100-26-SEP	19	3	3	11
KXBTCMAX150-25-26APR30-149999.99	45	3	22	30
KXBTCMAX150-25-26FEB28-149999.99	81	10	18	50
KXBTCMAX150-25-26JAN31-149999.99	67	14	20	37
KXBTCMAX150-25-26MAR31-149999.99	62	5	37	44
KXBTCMAX150-25-26MAY31-149999.99	56	4	12	30
KXBTCMAX150-25-DEC31-149999.99	129	23	23	112
KXBTCMAXY-25-DEC31-129999.99	88	5	36	67
KXBTCMAXY-25-DEC31-139999.99	77	4	45	61

EDA


Jump Signals: KXBTCMAXY-25-DEC31-129999.99
D1 — 88 signals
D2 — 5 signals
prob_mid
1.0
0.8
0.6
0.4
0.2
0.0

Analysis

- Conducted lead-lag analysis for all lags -60 to 60
- What was BTC return T minutes after a signal?
- Definition Analysis
  - D4 had higher correlation, less varied "best lags", performed best in Granger
  - D2 and D3 barely had predictive power
- Cross correlation results
  - D4: best lag times are between 23 to 30 minutes
  - D1: best lag times range from 6 to 45 minutes

Analysis

- Granger causality: does prediction market history contain information about BTC's future that BTC's history doesn't already contain/explain?
- Confirms that they're not just reacting to the same underlying cause
- For top markets, p-value ~ 0, extremely significant
- Granger causality applied to entire dataset, so statistically significant relationship between ΔP and BTC **confirmed.**
- Tested on 60 market/definition pairs, 43 pairs significant at $p<0.05$, 25 after Bonferroni correction

Information edge is statistically significant, how do we convert this into a tradeable strategy?

Event Case Study

- Long BTC after upward jump, short after downward jump
- Direction-adjusted Cumulative Abnormal Return, 240-min window, 10bp cost assumption

Definition	N Events	Peak Net CAR	p-value	Pass?
D1	304	−0.12%	0.17	✗ FAIL
D2	44	+0.005%	0.54	✗ FAIL
D3	18	+0.26%	0.19	✗ FAIL
D4	170	−0.13%	0.35	✗ FAIL

- All 4 definitions fail in top 5 markets. No statistically significant edge when entering at T+1
- Does not invalidate Granger, just confirms the edge lives in the 0-1 minute timeframe

Backtest

- Strategy: enter BTC long/short at T+1 after signal, hold H minutes, exit on stop-loss
- Costs: 0.075% commission (buy+sell) + 0.025% slippage + 1.5× vol stop-loss
- Data split: 60/20/20 chronological by events

Def	H=10 Sharpe	H=30 Sharpe	H=60 Sharpe	H=240 Sharpe	H=240 Max DD	H=240 Trades
D1	-2.30	-1.92	-1.31	+0.37	16.6%	30
D2	-1.79	-0.71	-0.34	+1.29	5.1%	44
D3	-1.88	-0.58	+0.03	+1.21	7.9%	65
D4	-2.36	-1.65	-0.58	+0.94	9.1%	66

Extensions

- **Sub-minute data**: edge lies here!
  - Scrape 10s Kalshi data over next month
- **Extend to other assets**
  - Run pipeline on ETH and SOL
- **Regime segmentation**
  - Check if lag differs under different market regimes
  - Combine strategy to capitalize on immediate edge + slow drift across market types
- Paper trade once results cross threshold metrics